AIMORA SYSTEM ARCHITECTURE SPECIFICATION

Institutional Feature, Workflow & Logic Manual

Comprehensive Blueprint of Faculty (CDC Director) and Student (Candidate) Modules

Platform: AIMORA Placement & Career Suite	**Scope:** Student & Faculty Consoles	**Focus:** Deterministic Scoring, Gates, SerpApi & AI Workflows	**Notice:** **Red Items** represent Critical Core Features

AIMORA is a next-generation Career Intelligence and Placement Operating System tailored for higher-education engineering institutions. It solves the critical disconnect between student self-reported skills, institutional placement eligibility gates, and real-time industry employer expectations. The platform enforces deterministic skill measurement, transparent proof-of-work validation, and automated remediation without synthetic placements.

🎓 Part 1: Student Module (Candidate Career Console)

The Student Module empowers candidates to progress along industry-verified learning curves, analyze and optimize resumes against job descriptions, practice with an AI Bar-Raiser mock interviewer, and unlock campus placement drives through verified competency milestones.

Student Feature Matrix

FEATURE	FUNCTIONAL PURPOSE	CLASSIFICATION
Deterministic Placement Readiness Engine CRITICAL	Computes objective, weighted placement readiness score based on verified competencies vs. industry target profiles.	**Core Algorithmic Gate**
Proof-of-Work Studio & Verification CRITICAL	Inspects and attaches verifiable artifact evidence (GitHub repos, assessment IDs, faculty approvals).	**Anti-Goodhart Verification**
Interactive Skill Tech Tree CRITICAL	Interactive visual dependency tree mapping prerequisite knowledge nodes across foundational to advanced stacks.	**Visual Learning Roadmap**
Multi-Format Profile Ingestion CRITICAL	Ingests student profiles via CSV, JSON, or PDF using a 30+ term canonical synonym lexicon and Gemini fallback parser.	**Data Ingestion Pipeline**
Target Role Selector & Dynamic Roadmap	Generates 30-day milestone phases, diagnostic quizzes, and prioritized gaps tailored to selected target roles.	Adaptive Pathway Planner
Instant Skill Verification Quizzes	3-question diagnostic assessments (SQL, Docker, Redis) providing rapid, verifiable proficiency updates.	Competency Assessment
Multi-Dimensional ATS Alignment Scoring CRITICAL	Evaluates resumes across Keyword Relevance, Evidence & Proof, and Bullet Clarity metrics in real time.	**Career Asset Intelligence**
One-Click Diff Application Engine CRITICAL	Side-by-side visual diff editor applying AI-recommended bullet point enhancements instantly to working drafts.	**Interactive Document Studio**
AI Bullet Enhancer & Cover Letter Generator	Applies Google X-Y-Z formula to rewrite resume bullets and crafts personalized job application cover letters.	AI Generation Suite
LinkedIn Profile Optimizer	Generates punchy About summaries, role-targeted headlines, and top skill endorsements for networking.	Personal Branding Suite
SerpApi Real-Time Live Job Radar CRITICAL	Queries Google Jobs API via SerpApi in real-time, retrieving live roles, company details, salaries, and posting ages.	**Live Market Intelligence**
Dynamic Candidate-to-Job Match Index CRITICAL	Computes real-time compatibility % by cross-referencing candidate skills with job specifications and partner weightings.	**Matching Engine**
Market Skill Demand Intelligence	Tracks market demand velocity across modern tech stacks to guide student skill acquisition priorities.	Market Analytics
Institutional Eligibility Gating Engine CRITICAL	Evaluates minimum CGPA, active backlogs, and verified competencies to grant or deny drive registration.	**Institutional Compliance**
Prerequisite Lab Verification Lock (#DCK-101) CRITICAL	Blocks high-tier corporate placement drives until specific mandatory technical labs are verified.	**Prerequisite Gating**
Reactive Multimodal Interviewer Avatar CRITICAL	3D/Audio simulated interviewer displaying responsive lip-sync, thinking, speaking, and listening states.	**Multimodal Simulation**

FEATURE	FUNCTIONAL PURPOSE	CLASSIFICATION
Real-Time Gemini Bar-Raiser Engine CRITICAL	AI interview conductor tailoring Technical and HR behavioral challenges with strict pushback on weak answers.	**Generative Bar Raiser**
Speech-to-Text & Text-to-Speech Integration CRITICAL	Voice-driven interview room using Web Speech Recognition and Web Speech Synthesis for hands-free practice.	**Voice Interface**
Automated CDC Scorecard Generation CRITICAL	Generates formal interview scorecards with Technical Depth, Problem Solving, STAR Compliance, and Communication.	**Diagnostic Evaluation**
Deterministic Semester Skill Verification CRITICAL	Semester-wise examinations (Semesters 3-8) directly mapping university course codes to verified competencies.	**Curriculum Seam Testing**
Student Career Copilot	Persistent conversational AI tutor recommending study tasks, launching quizzes, and explaining roadmap logic.	Conversational Assistant

Student Workflows & Underlying Logics

1. Deterministic Readiness Scoring Workflow & Mathematical Logic

CORE LOGIC

1. Student completes verification quiz, semester exam, or imports proof-of-work repository.
2. System validates evidence provenance (self-reported, verified assessment, or faculty endorsed).
3. System runs deterministic ratio evaluation for every competency in the candidate's target stack.
4. Placement readiness gauge updates instantly; drives re-evaluate candidate eligibility.

Mathematical Formula:

$$\text{Readiness } \% = \left[\Sigma(w_i \times \text{Ratio}_i) \,/\, \Sigma\, w_i \right] \times 100$$

where $\text{Ratio}_i = \min(1.0,\ \text{Proficiency}_i \,/\, \max(1.0,\ \text{TargetLevel}_i))$

Stack Weights (w_i): Python: 24 | Git: 22 | System Design: 21.6 | SQL: 12 | Docker: 10 | Redis: 8 | Kafka: 2.4

Null-Safety Invariant: Unassessed skills are stored as `null` (never 0.0). Null skills contribute 0 earned score but are flagged as "Unverified Mentions" rather than failed competencies.

2. Multi-Dimensional ATS Resume Scoring & Diff Application

CORE LOGIC

1. Student inputs or edits working resume draft within the side-by-side workspace.
2. Engine runs lexical and semantic scans against selected target role requirements.
3. Score is decomposed into three distinct sub-indices: Keyword Relevance, Evidence & Proof, and Clarity.
4. Student reviews AI-recommended diff suggestions and applies them with one click.

Alignment Formulas:

$$\text{Relevance Score} = \min(100,\ 74 + \Delta_{\text{keyword_matches}})$$

$$\text{Evidence Score} = \min(100,\ 62 + \Delta_{\text{metrics_and_repos}})$$

$$\text{Clarity Score} = \min(100,\ 88 + \Delta_{\text{action_verbs}})$$

$$\text{Overall ATS Alignment} = \text{Round}((\text{Relevance Score} + \text{Evidence Score} + \text{Clarity Score}) \,/\, 3)$$

3. AI Bar-Raiser Multimodal Mock Interview Loop

CORE LOGIC

1. Student configures target company (e.g., Amazon SDE-1) and mode (Technical Bar Raiser vs. HR Behavioral).
2. Web Speech Recognition captures candidate's vocal answer and streams transcribed text to editor.
3. System validates answer sufficiency (detecting dismissive or trivial responses < 15 chars).
4. Gemini 1.5/2.0 evaluates depth, provides conversational pushback, and prompts the next stage.
5. Web Speech Synthesis renders realistic voice output while the Avatar renders synced lip animations.
6. Post-interview scorecard compiles scores across Technical Depth (40%), STAR Methodology (30%), Problem Solving (20%), and Communication (10%).

🏛️ Part 2: Faculty & CDC Module (Placement Governance Console)

The Faculty / Placement Director Module enables department heads, career development cells (CDC), and faculty advisors to monitor cohort readiness, identify institutional curriculum deficits, enforce corporate eligibility rules, and automate remediation bootcamps.

Faculty Feature Matrix

FEATURE	FUNCTIONAL PURPOSE	CLASSIFICATION
Executive Placement KPI Dashboard	Real-time macro dashboard tracking Overall Placement %, Cohort Readiness Avg, Active Drives, and At-Risk count.	Executive Overview
Cohort Skill Deficit Heatmap CRITICAL	Cross-sectional matrix displaying average competency scores and null/unverified rates across departments and sections.	**Institutional Analytics**
Automated Risk Flagging & Alert Feed CRITICAL	Continuous monitoring feed flagging candidates with CGPA drops, prerequisite deficits, or missing verifications.	**Early Intervention**
Deep Student Dossier & Proof Verification CRITICAL	Detailed inspection view of individual student competencies, linked GitHub repos, project evidence, and verified history.	**Candidate Auditing**
Faculty Eligibility Override Governance CRITICAL	Controlled administrative override allowing faculty to unlock restricted drives with mandatory justification.	**Administrative Governance**
Dynamic Benchmark Deficit Detector CRITICAL	Interactive threshold slider calculating in real time how many students fail to meet target competency bars.	**Cohort Diagnostics**
One-Click Batch Remediation Dispatcher CRITICAL	Automated intervention scheduler that creates mandatory workshops and dispatches tasks to deficit cohorts.	**Automated Remediation**
Configurable Drive Rule Engine & Cohort Simulator CRITICAL	Interactive rule configurator allowing directors to adjust drive CGPA/readiness criteria and simulate pool size.	**Drive Governance**
Tamper-Evident Override Audit Trail CRITICAL	Immutable record of all faculty endorsements, gate exceptions, and status overrides exportable to signed CSV.	**Compliance & Audit**
Faculty Endorsement & Feedback Dispatcher CRITICAL	Two-way feedback console allowing faculty to endorse student portfolio claims or request structured revisions.	**Mentorship Operations**
NAAC / NBA Accreditation & Compliance Engine CRITICAL	Report generator mapping institutional competency scores directly to NAAC Criterion 5 and NBA OBE metrics.	**Accreditation Suite**
FERPA / DPDP Privacy Audit Export CRITICAL	Generates privacy-compliant candidate packages with anonymized identifiers for external corporate recruiters.	**Data Privacy & Protection**
Pre-Flight Integrity Validator & Synonym Resolver CRITICAL	Scans imported class spreadsheets for malformed data, GPA scale mismatches, and header naming discrepancies.	**Data Hygiene Engine**
Configurable Duplicate Reconciliation Engine CRITICAL	Resolves student ID conflicts using Merge & Update, Reject Duplicate, or Create Aliased Identity policies.	**Database Reconciliation**

FEATURE	FUNCTIONAL PURPOSE	CLASSIFICATION
AI Cohort Analytics & Instant Nudge Broadcaster CRITICAL	Faculty mode Career Copilot parsing natural language queries into automated workshops and batch student nudges.	**AI Administrative Assistant**

Faculty Workflows & Underlying Logics

1. Cohort Skill Deficit Heatmap Logic & Threshold Rules

CORE LOGIC

1. Director selects Department (e.g. CSE), Batch (2026), and Section (e.g. CSE-A, CSE-B).
2. System aggregates student competency records and computes section averages and null percentages.
3. Color-coded risk classifications are rendered automatically based on institutional benchmarks.

Heatmap Classification Rules:

$$\text{Average Proficiency} = \Sigma\ \text{Proficiency}_{valid} \ /\ N_{valid}$$

$$\text{Null / Unverified Rate} = (N_{null} \ /\ N_{total}) \times 100\%$$

Alert States:

- **Critical Deficit (Red):** Average $< 2.8 / 5.0$ OR Null Rate $\geq 30\%$
- **Moderate Warning (Yellow):** Average $< 3.5 / 5.0$ OR Null Rate $\geq 15\%$
- **Optimal Attainment (Green):** Average $\geq 3.5 / 5.0$ AND Null Rate $< 15\%$

2. Drive Rule Configuration & Live Cohort Simulation

CORE LOGIC

1. Director selects an upcoming campus drive (e.g., CloudScale Super Dream 2026).
2. Director adjusts gating parameters (Min CGPA: 3.50, Min Readiness: 85%, Mandatory Docker Gate: ON).
3. System simulates cohort eligibility dynamically, reporting eligible vs. remediation counts in real-time.
4. Rules are saved and enforced across student candidate views instantly.

Candidate Gate Condition:

$$\text{Eligible} = (\text{Student.CGPA} \geq \text{Rule.MinGPA}) \wedge (\text{Student.Readiness} \geq \text{Rule.MinReadiness}) \wedge (\text{DockerCompleted} == \text{True} \vee \text{OverrideActive})$$

Cohort Simulation Model:

$$\text{Simulated Eligible} = \text{Round}(\ \text{BaseEligible} \times \text{ReadinessFactor} \times \text{GPAFactor}\ )$$

where ReadinessFactor = 1.4 (if Readiness $< 80\%$), 1.0 (80-85%), 0.7 (if $> 85\%$)

GPAFactor = 0.6 (if Min GPA > 3.6), 1.0 (otherwise)

3. Batch Remediation & Automated Intervention Pipeline

CORE LOGIC

1. Faculty opens Cohort Skill Analytics and selects targeted skill deficit (e.g., Docker).
2. Interactive benchmark threshold slider is adjusted to desired competency level (e.g., 3.5).
3. Engine isolates all students with scores below threshold or missing verification.
4. Faculty triggers "Dispatch Batch Workshop" → Workshop calendar event created & notifications sent.

Target Deficit Filter:

$$\text{Deficit Students} = \{\ s \in \text{Cohort} \mid s.\text{skills}[\text{selectedSkill}] == \text{null} \vee s.\text{skills}[\text{selectedSkill}] < \text{Threshold}\ \}$$

Dispatched Action: Automatic insertion into Faculty Workshop Schedule and candidate Roadmaps.

🔄 Part 3: Cross-Module Interaction & Data Flow Architecture

The Student and Faculty modules operate as an interconnected closed-loop system. Student self-improvement directly informs institutional placement health, while institutional gates and faculty interventions drive structured student preparation.

Inter-Module Feedback Loop

INTERACTION SEAM	STUDENT MODULE ACTION	FACULTY MODULE COUNTERPART
Competency Verification	Student completes Semester Assessment or Docker Quiz (#DCK-101), elevating skill score to 4.0+.	Section heatmap automatically recalculates, clearing critical red flags and updating cohort readiness KPIs.
Placement Gating	Student attempts to apply for CloudScale Super Dream drive; blocked by Gated Prerequisite alert.	Director sets prerequisite lab rules or inspects student dossier to grant authorized faculty override.
Remediation Pipeline	Student receives automated roadmap task: "Mandatory Remediation Workshop: Docker Containerization".	Faculty identifies 24 deficit students via Benchmark Slider and dispatches batch remediation workshop.
Portfolio Endorsement	Student uploads GitHub proof-of-work repository or requests formal portfolio review.	Faculty inspects evidence in Reviews & Training console, approving claim with an official institutional seal.
Accreditation Reporting	Student cohort progresses through semester assessments, recording verifiable skill proficiencies.	CDC Director exports NAAC Criterion 5 and NBA OBE attainment reports with verifiable audit trails.

Core Institutional Invariants

- **Null-Safety Invariant:** Unknown competencies are represented strictly as `null`, never `0.0`, preventing false negative penalization during initial imports.
- **Placement Gate Integrity:** Drive applications require deterministic validation of CGPA, verified prerequisite skills, and zero active disciplinary flags unless overridden with an audited justification.
- **Proof-of-Work Provenance:** Every competency score in the system is tied to an explicit provenance tag: `self-reported`, `verified_assessment`, `github_repo`, or `faculty_endorsed`.
- **FERPA / DPDP Privacy Boundary:** Student identifiable data is never exposed to external search APIs (such as SerpApi) or third-party recruiters without explicit compliance filtering.